

Scalability & Future Plan

Date	9 July 2026
Project Name	Rising Waters - Real-time Flood Prediction and Alert System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-node localized deployment environment for backend ML inference models.	System encounters bottleneck latency spikes under high parallel API query loads.	High
2	Relational schema structure storing continuously updating spatial-temporal sensor variables.	Slower analytical search query response times as historical log volume accumulates.	Medium
3	Single region SMS microservice routing for all localized mobile client updates.	Network latency delays or dropped alert distributions across wider external borders.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles up to 50 concurrent active real-time dashboard sessions.	Transition to a containerized cluster behind an Elastic Load Balancer to dynamically autoscale instances.
2	Data Storage	Standard relational table files storing periodic log telemetry.	Migrate high-velocity time-series sensor inputs into a dedicated distributed NoSQL engine.
3	Performance	On-demand model execution logic computes inputs linearly.	Implement a Redis caching cluster layer for frequent tracking queries and run distributed inference arrays.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	Security	Basic token authentication mechanism guarding access routes.	Introduce multi-factor authentication protocols, granular RBAC matrices, and full TLS-level asset encryption.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-modal satellite imagery ingestion pipeline integration.	3 Months	Significantly improves precision metrics in low-density sensor zones.
Phase 3	Edge deployment framework tailored for specialized mesh networks.	6 Months	Ensures persistent warning updates even during total grid failure events.
Phase 4	Cross-agency automated emergency response API integration.	9 Months	Reduces disaster response latency down to sub-minute operational cycles.